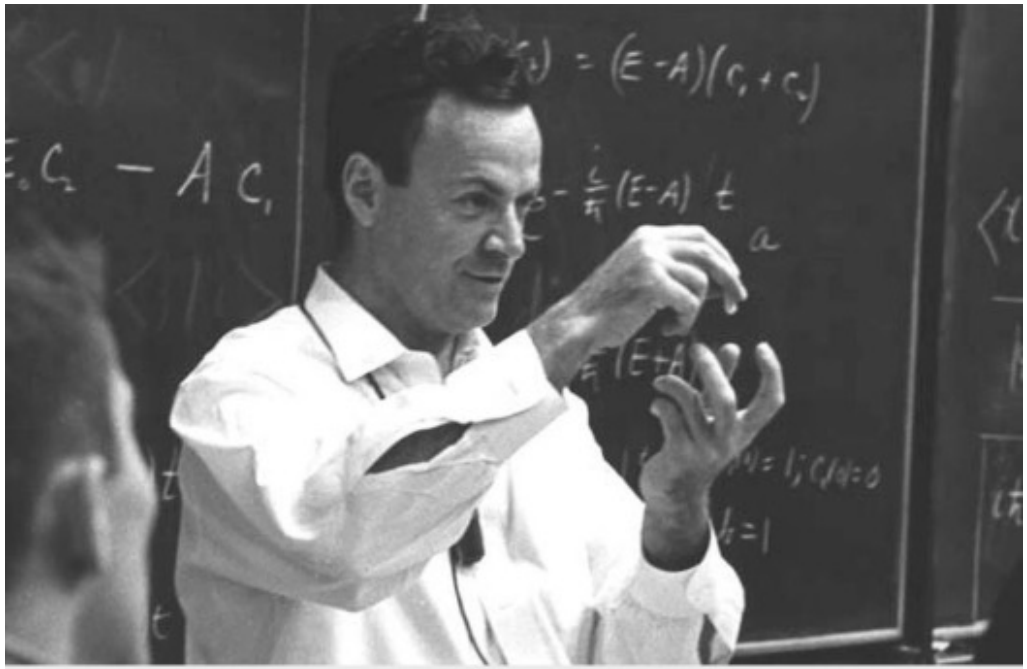
Object Detection & Recognition using YOLO

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October 6, 2026

Prof. Dr. Richard P. Feynman



Knowledge isn't free.
You have to pay attention.

- American theoretical physicist
- Nobel Prize in Physics, 1965 (Shared with Julian Schwinger and Shin'ichirō Tomonaga)

คิดล้ำ นำวิจัย พร้อมใช้ในอุตสาหกรรม





(ตรงต่อเวลา รู้หน้าที่ มีวินัย)

Chapter Goals

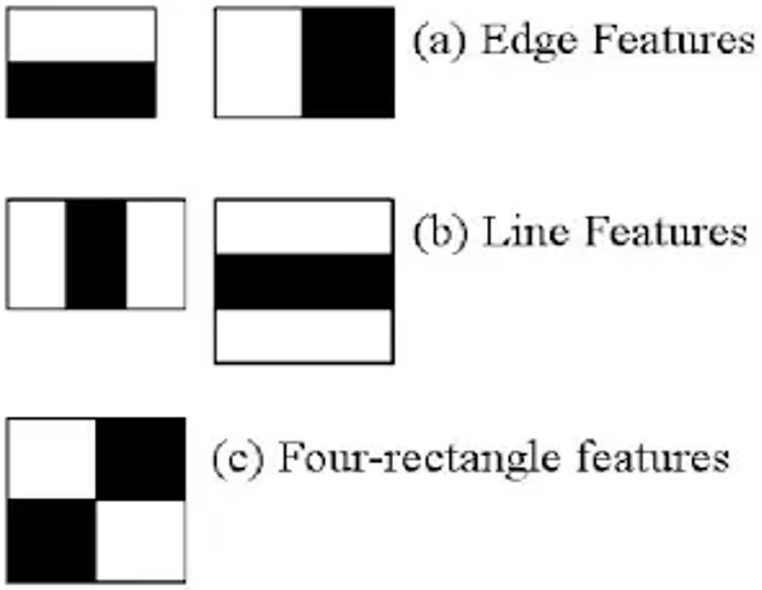
- To understand basic knowledge of YOLO
- To detect objects with single stage (Single-shot object detection) using YOLO

Topics:

- What is YOLO?
- Why the YOLO algorithm is important
- How the YOLO algorithm works
- Object detection using YOLO

Recap: Object Detection using Haar Like Feature (The Viola–Jones Algorithm)

- *Change* the image to *grayscale*
- *Haar-like features* are *scalable*, rectangular frames that are used to *compare* how pixels relate *to each other*; specifically *how dark* one is to the other.
- The *white bars* represent pixels that contain parts of an image that are *closer to the light* source, and would therefore be "*whiter*" on a grayscale image.
- The *black bars* are the opposite.



Recap: The Viola–Jones Algorithm

Important note:

- Haar-like features are scalable.
- Then, *Edge features*, it could be 1x2, 100x200, or even 400x50 pixels. It doesn't matter!
- The only *dimension* they *can't be* is 1x1 pixels.
  - So keep that in the back of your mind as we continue.

Recap: The Viola–Jones Algorithm: Face detection

Two objects: *forehead* and the *eyes/eyebrows*.

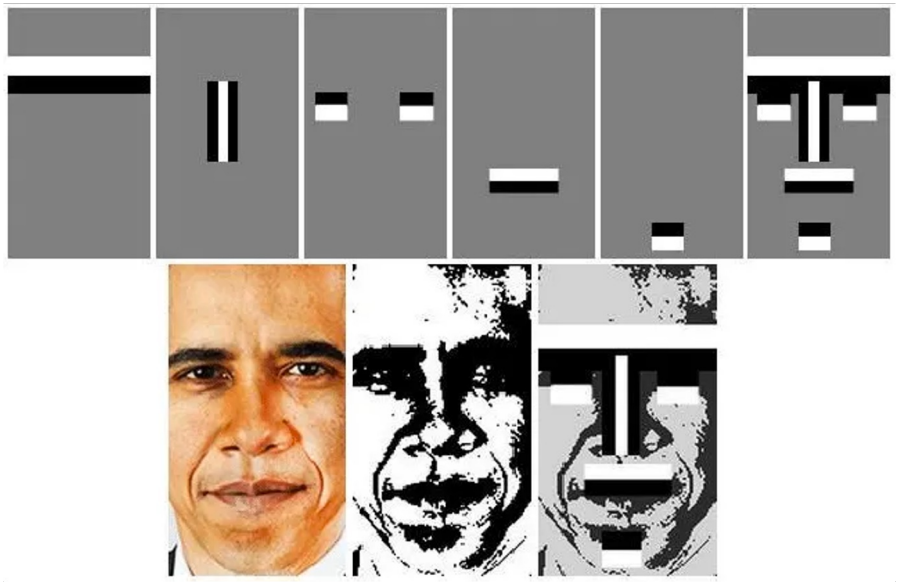
- The *forehead* is an exposed, flat surface of the face. This allows it to reflect *more light*, so it is often "*lighter*".
- *Eyebrows* are generally *darker*.
- The algorithm would read the *lighter shade* of the *forehead* and the transition to the *darker eyebrows* as an "*edge*" boundary.

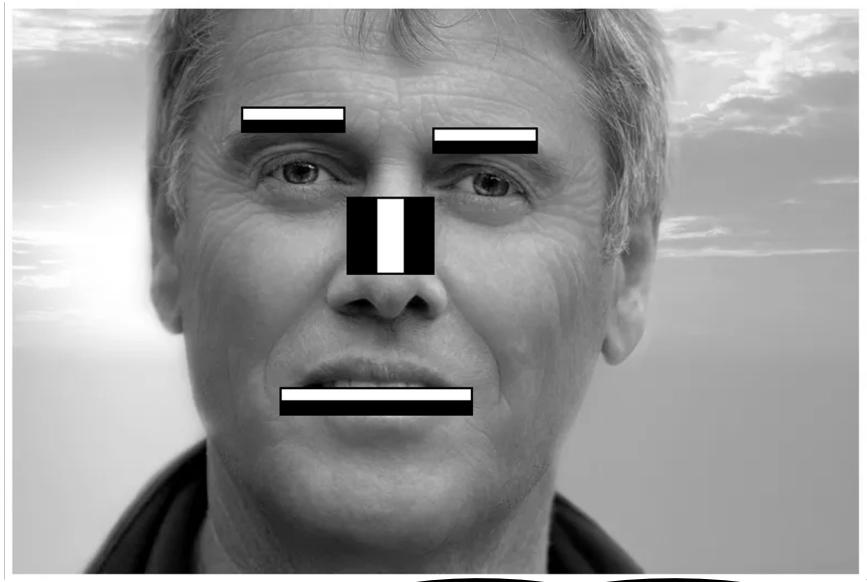
Recap: The Viola–Jones Algorithm: Line Feature

The pattern can go *white-black-white*, or *black-white-black* (like an *Oreo*)

- Pattern of Nose is *flat-not flat-flat*
- Pattern of Eye is *white-dark (iris)-white*

Recap: The Viola–Jones Algorithm: Application





Accuracy &
Bounding box

Can we improve the performance?

What is YOLO?

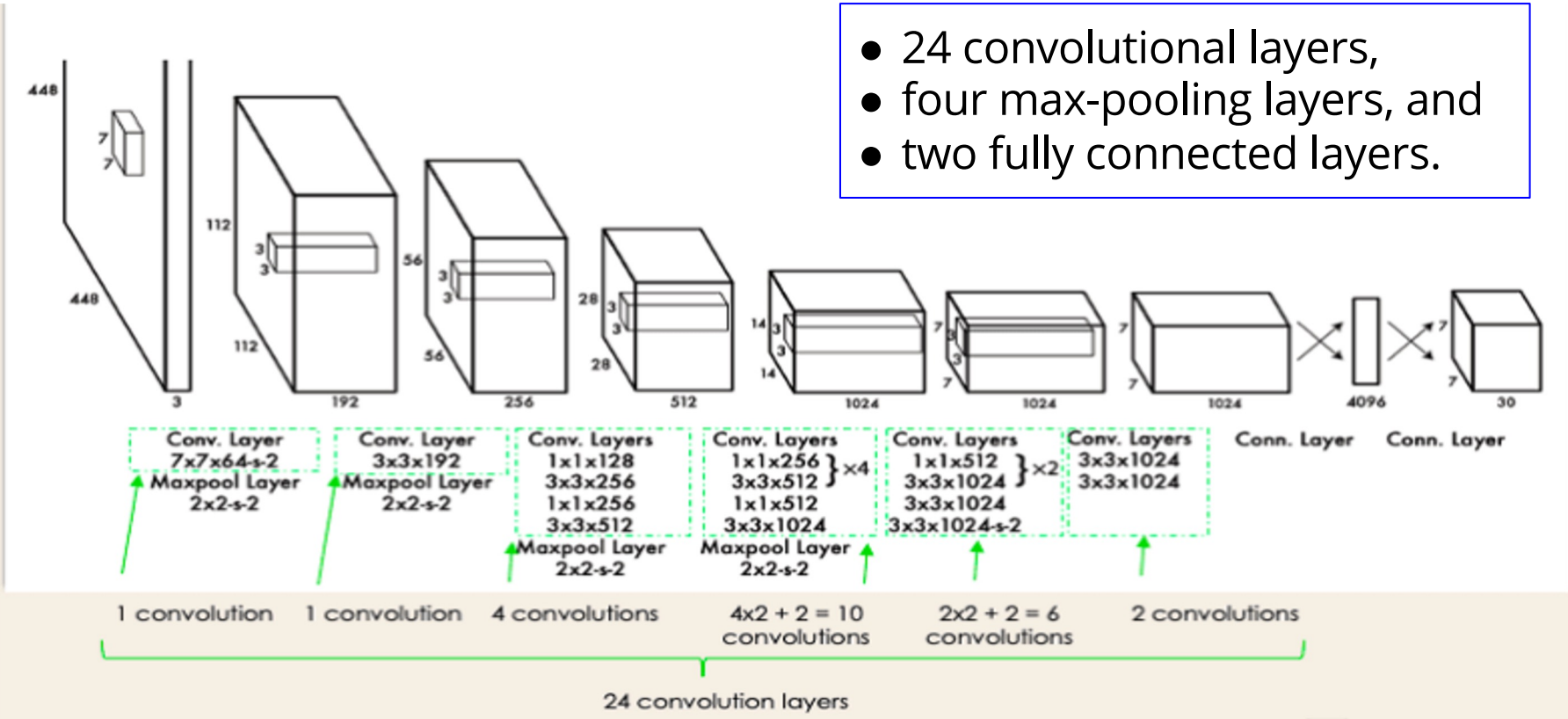
You Only Look Once: *YOLO* is an algorithm that uses *convolutional neural networks* to provide real-time object detection.

By

- Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi, (2015), in their famous research paper “You Only Look Once: Unified, Real-Time Object Detection”.

YOLO Architecture

- 24 convolutional layers,
- four max-pooling layers, and
- two fully connected layers.



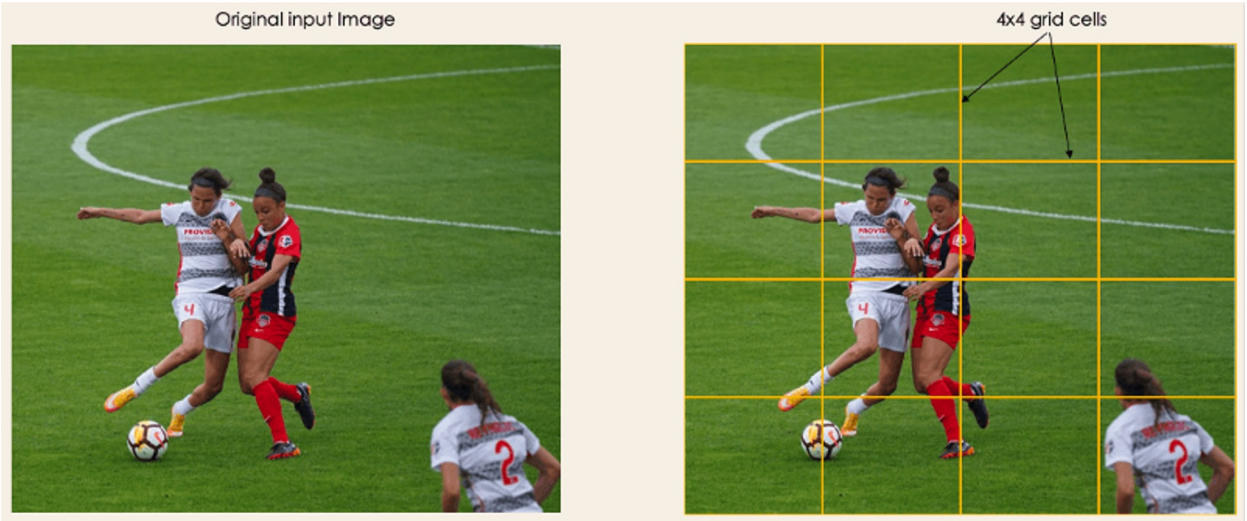
Resizes the input image into **448 x 448** before going through the convolutional network.

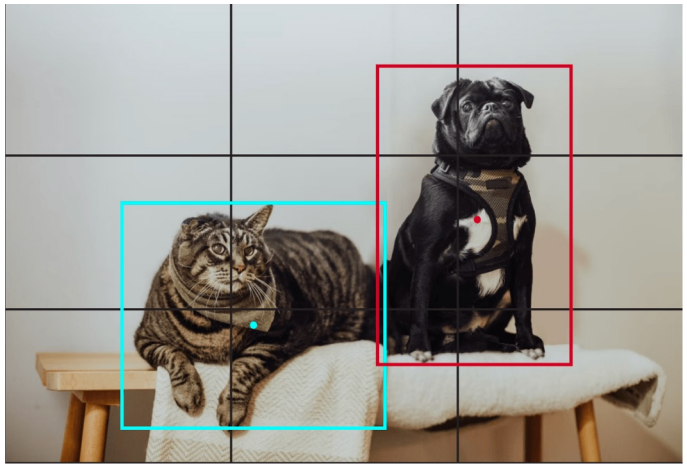
How does YOLO object detection work?

- Residual blocks
- Bounding box regression
- Intersection Over Unions or IOU for short
- Non-Maximum Suppression.

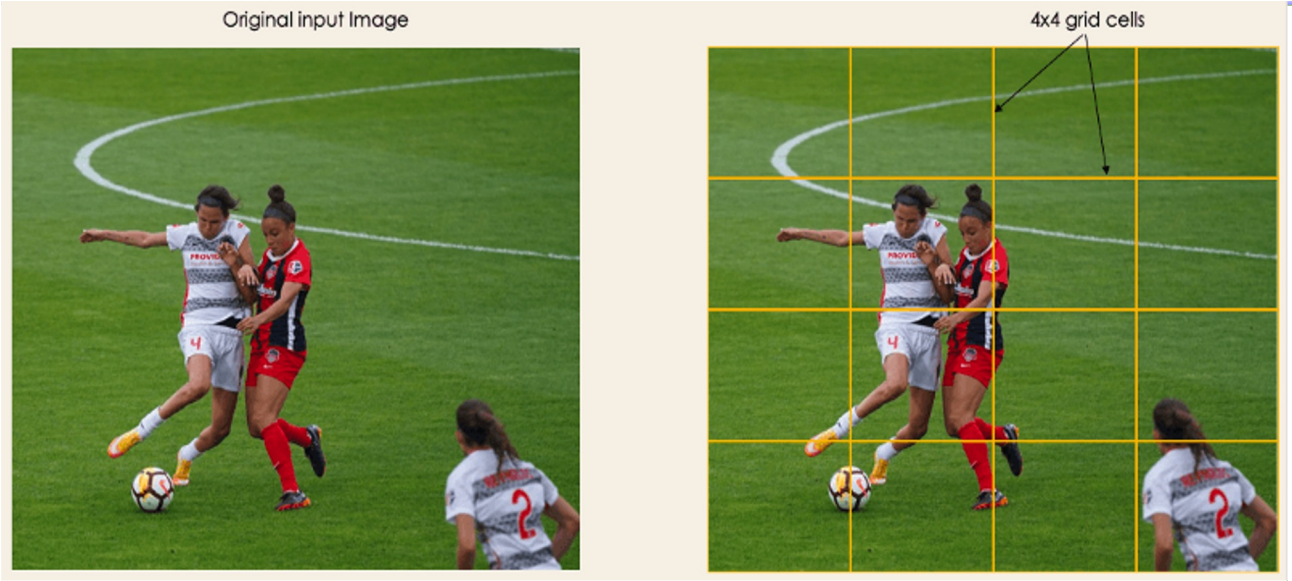
1-Residual blocks

- This *first step* starts by dividing the original image (A) into *NxN* grid cells of *equal shape*.
- Each *cell* in the grid is responsible for *localizing and predicting* the class of the object that it covers, along with the *probability/confidence value*.





Q and A: YOLO รู้ได้อย่างไรว่ามีวัตถุอยู่ใน Grid?



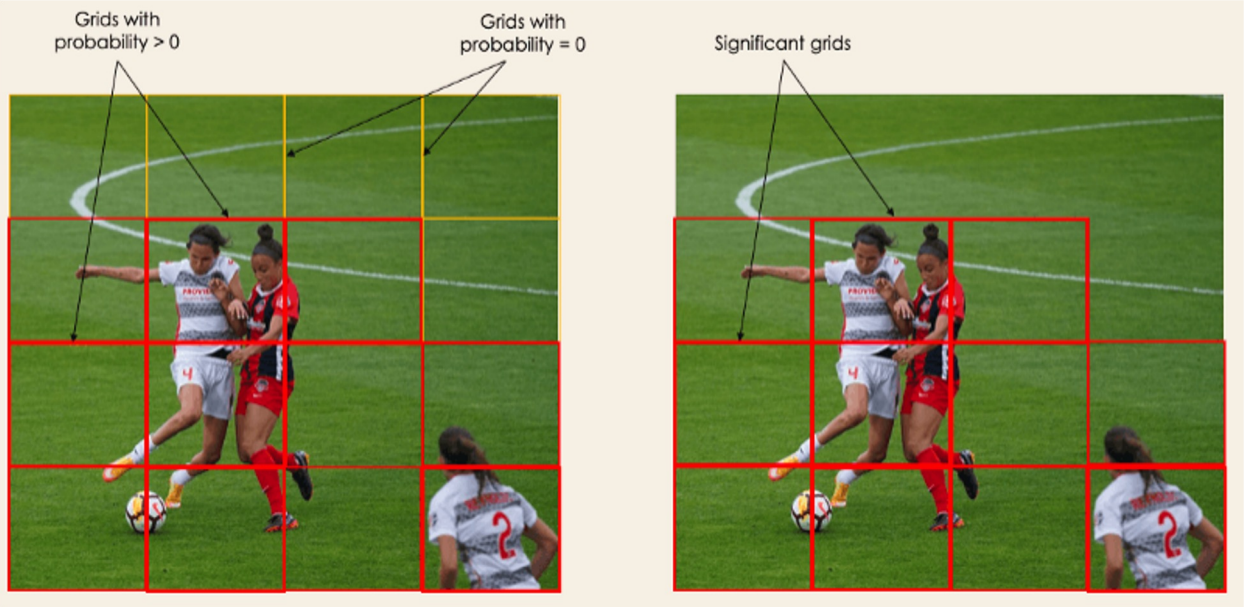
2- Bounding box regression

pc corresponds to the probability score of the grid containing an object.

- For instance, all the grids in red will have a probability score higher than zero.
- The image on the right is the simplified version since the probability of each yellow cell is zero (insignificant).

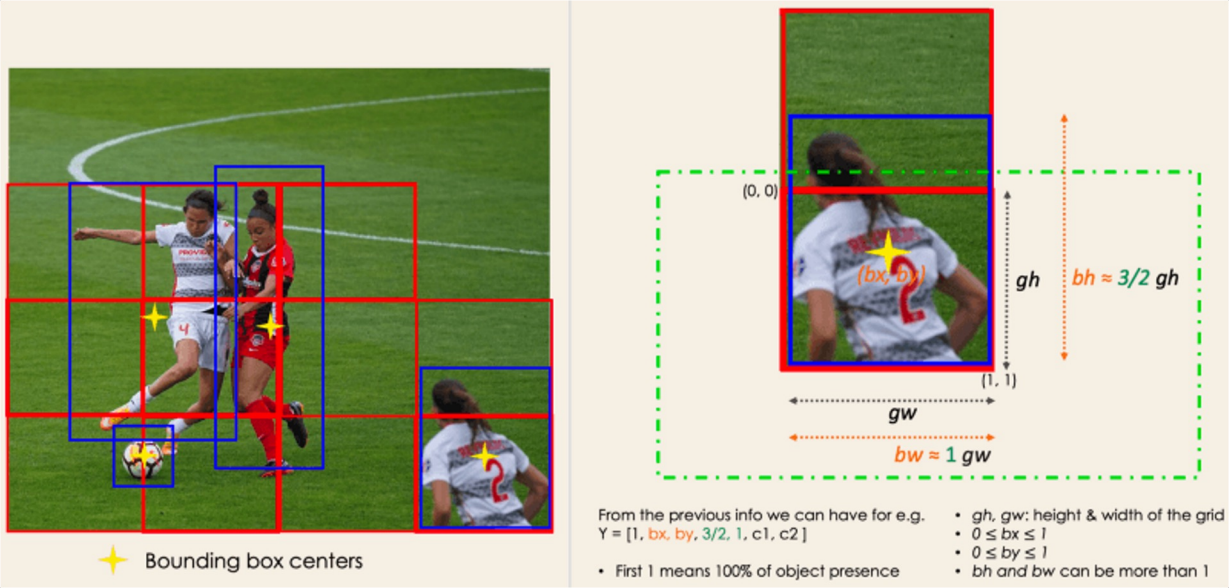
YOLO determines the attributes of these bounding boxes using a single regression module in the following format.

Y = [*pc*, bx, by, bh, bw, c1, c2]



2- Bounding box regression

- bx, by are the *x* and *y* coordinates of the center of the bounding box with respect to the enveloping grid cell.
- bh, bw correspond to the height and the width of the bounding box with respect to the enveloping grid cell.
- c1, c2 correspond to the two classes Player and Ball. We can have as many classes as your use case requires.



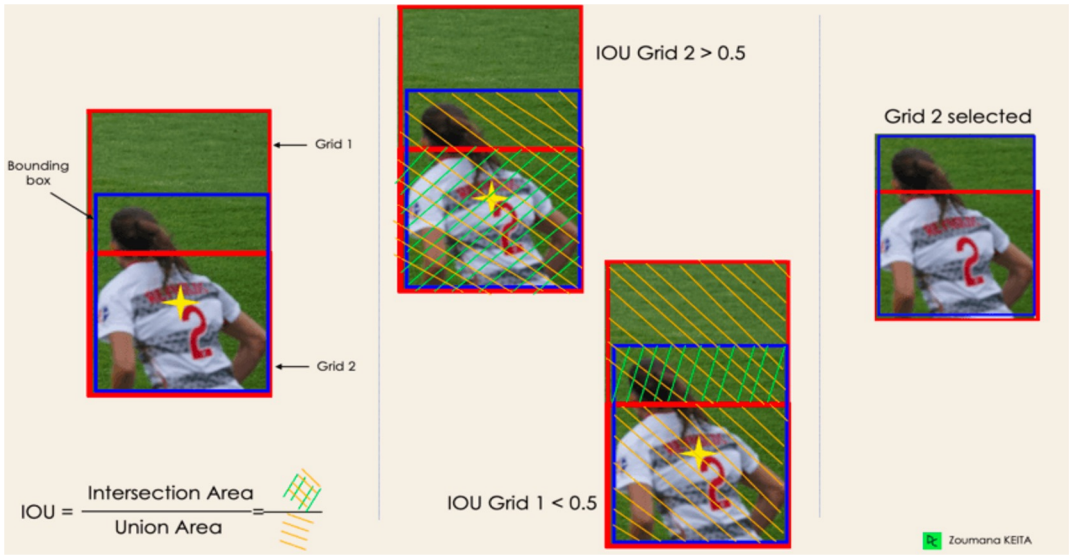
3- Intersection Over Unions or IOU

- A single object in an image can have multiple grid box candidates for prediction
- The goal of the *IOU* (a value between 0 and 1) is to discard such grid boxes to only keep those that are relevant
  - Defines its *IOU* selection threshold (*thr*)
  - Computes the *IOU* of each grid cell
  - If *IOU* < *thr*

    Ignore the grid cell

    Else

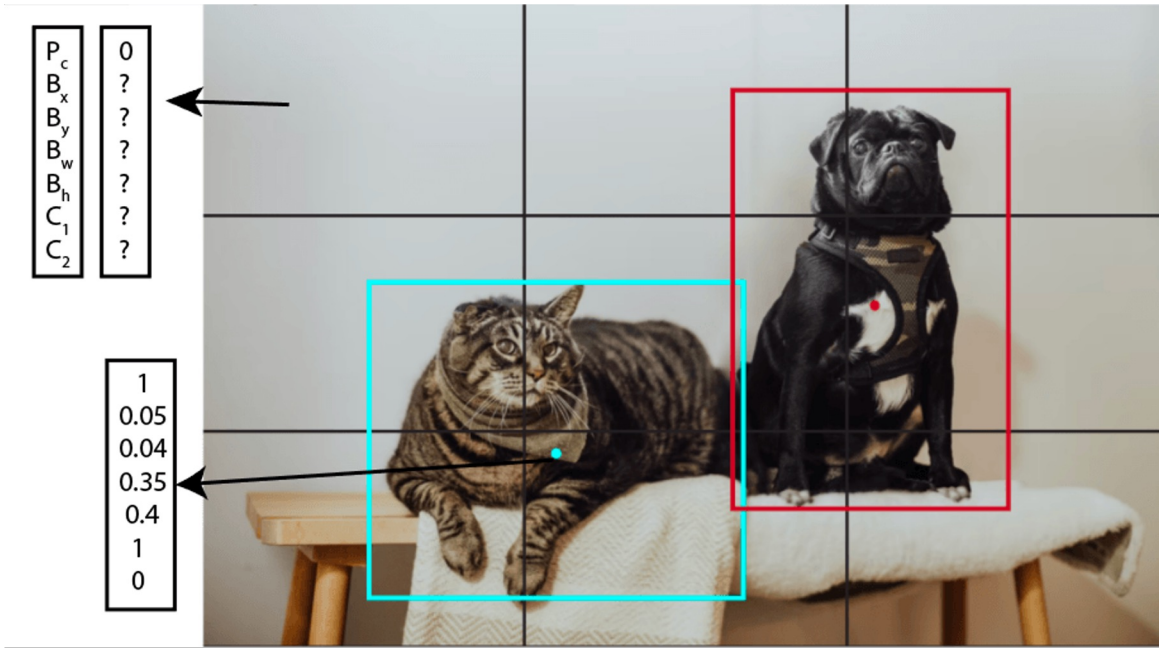
    Consider the grid cell



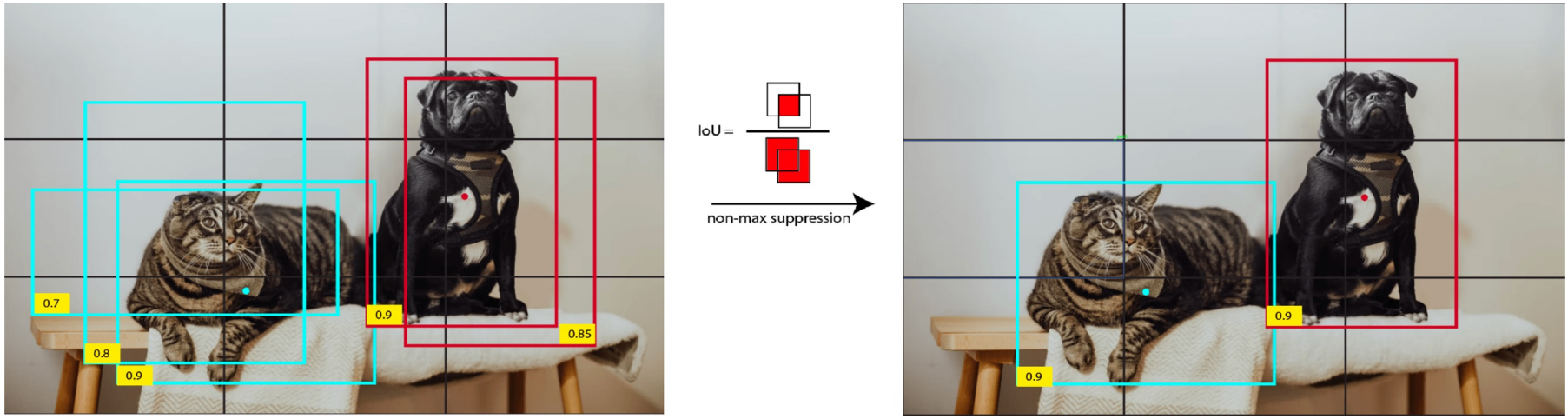
4- Non-Max Suppression or NMS

We could select only one box per class, that has the highest probability, but what if there are more objects of one class on the image.

- **Big Cat** is in more than one box.



$$IoU = \frac{\text{area of the intersection between } B_1 \text{ and } B_2}{\text{area of the union between } B_1 \text{ and } B_2},$$



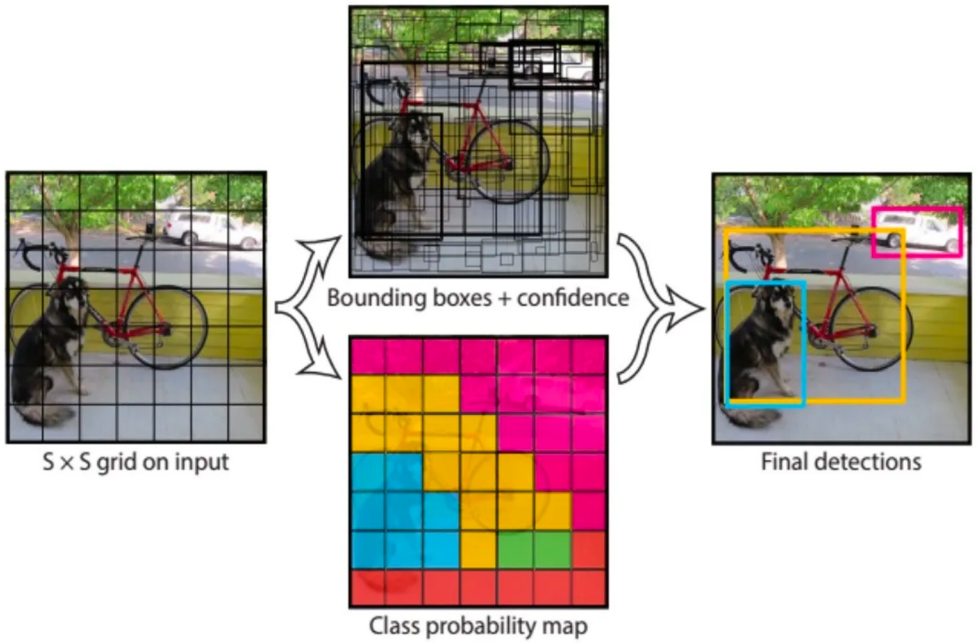
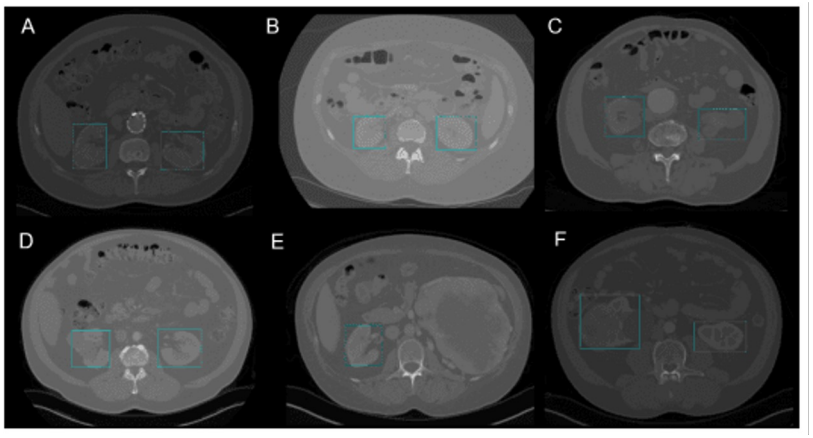


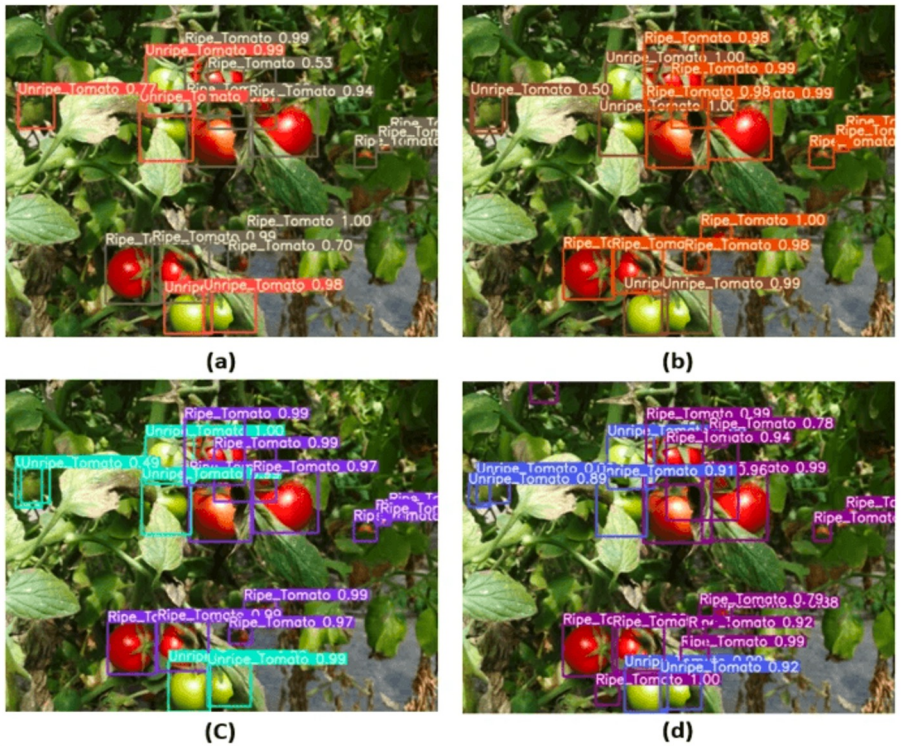
Figure 2: The Model. Our system models detection as a regression problem. It divides the image into an $S \times S$ grid and for each grid cell predicts B bounding boxes, confidence for those boxes, and C class probabilities. These predictions are encoded as an $S \times S \times (B * 5 + C)$ tensor.

Src: Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi, (2015), in their famous research paper "You Only Look Once: Unified, Real-Time Object Detection".

YOLO Applications



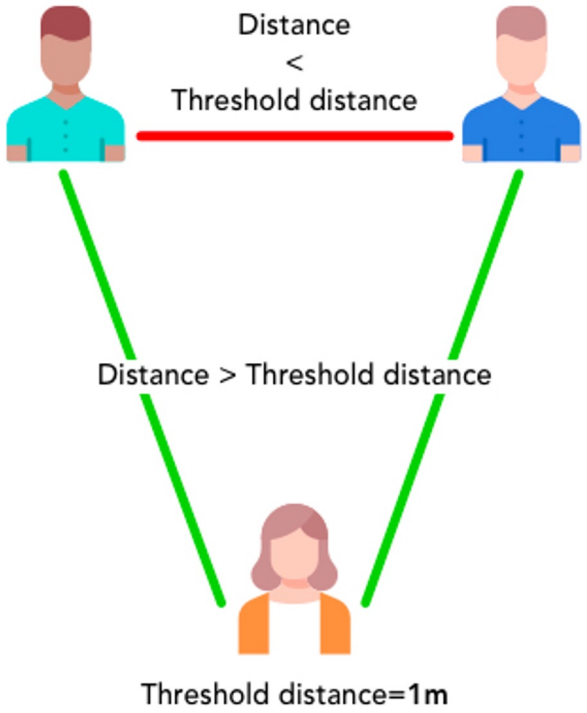
Healthcare: Kidney



Agriculture

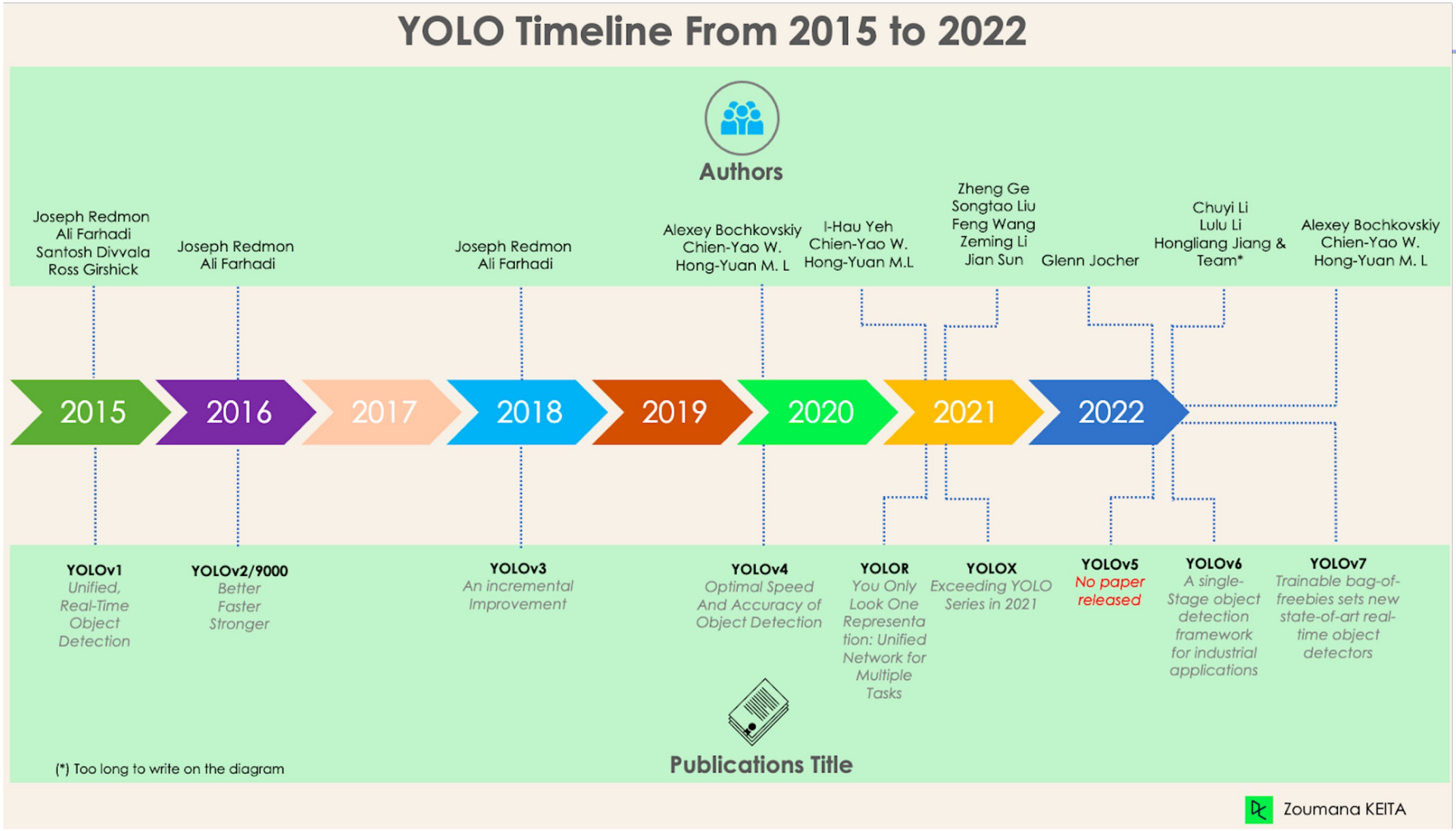
YOLO Application

YOLO in surveillance



What Makes YOLO Popular for Object Detection?

- Speed
- Detection accuracy
- Good generalization
- Open-source


YOLO Timeline From 2015 to 2022
Authors
Joseph Redmon
Ali Farhadi
Santosh Divvala
Ross Girshick
Joseph Redmon
Ali Farhadi
Joseph Redmon
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Alexey Bochkovskiy
Chien-Yao W.
Hong-Yuan M. L
I-Hau Yeh
Chien-Yao W.
Hong-Yuan M.L
Zheng Ge
Songtao Liu
Feng Wang
Zeming Li
Jian Sun
Glenn Jocher
Chuyi Li
Lulu Li
Hongliang Jiang &
Team*
Alexey Bochkovskiy
Chien-Yao W.
Hong-Yuan M. L
2015
2016
2017
2018
2019
2020
2021
2022
YOLOv1
Unified, Real-Time Object Detection
YOLOv2/9000
Better Faster Stronger
YOLOv3
An incremental Improvement
YOLOv4
Optimal Speed And Accuracy of Object Detection
YOLOR
You Only Look One Representation: Unified Network for Multiple Tasks
YOLOX
Exceeding YOLO Series in 2021
YOLOv5
No paper released
YOLOv6
A single-Stage object detection framework for industrial applications
YOLOv7
Trainable bag-of-freebies sets new state-of-art real-time object detectors
Publications Title
(*) Too long to write on the diagram
Zoumana KEITA

Term Project

Term Project

Write a program for object detection and recognition

- Create your own dataset
  - RayongMARCO project
- Propose CNN learning model
- Accuracy more than 90%

Rayong MARCO

ในการตรวจข้อสอบในปัจจุบันจำเป็นต้องใช้เครื่องตรวจ OMR / OCR โดยใช้การตรวจด้วยมือเปล่าซึ่งทำให้เกิดปัญหาต่างๆ ตามมา แต่ในปัจจุบันสัญญาณ Internet สามารถเข้าถึงได้ทุกพื้นที่ของประเทศไทย

“จะทำอย่างไรจึงจะสามารถใช้ประโยชน์จากสัญญาณ Internet เพื่อทดแทนการนำเข้าข้อมูลผ่าน Scanner ”

ปัญหาจากการใช้วิธีการตรวจข้อสอบแบบเก่า

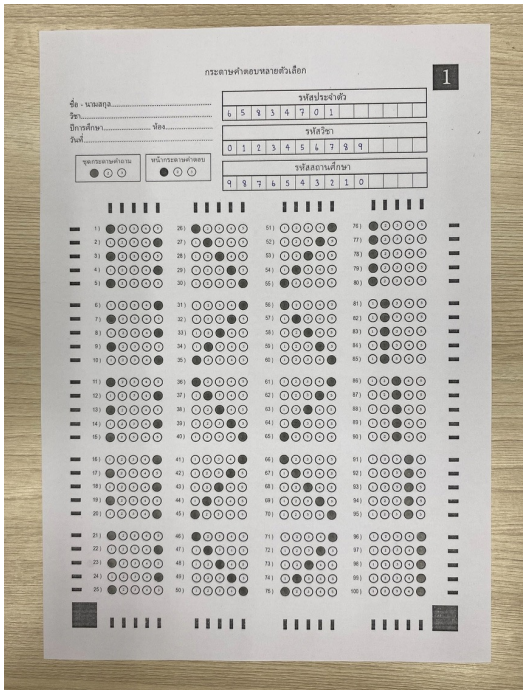
1. การเข้าถึงเครื่องตรวจข้อสอบ Scanner มีความลำบากในบางกรณี หรือบางโรงเรียน
2. ใช้เวลาในการตรวจหากตัดสินใจจะตรวจด้วยอาจารย์เองและอาจเกิดข้อผิดพลาดในการตรวจได้
3. ใช้เวลาในการตรวจหากตัดสินใจที่จะส่งชุดข้อสอบไปให้ผู้ที่มีส่วนเกี่ยวข้องช่วยตรวจข้อสอบ
4. สร้างความยุ่งยากให้ทุกภาคส่วน



ทำอย่างไรกับปัญหาเหล่านั้น?

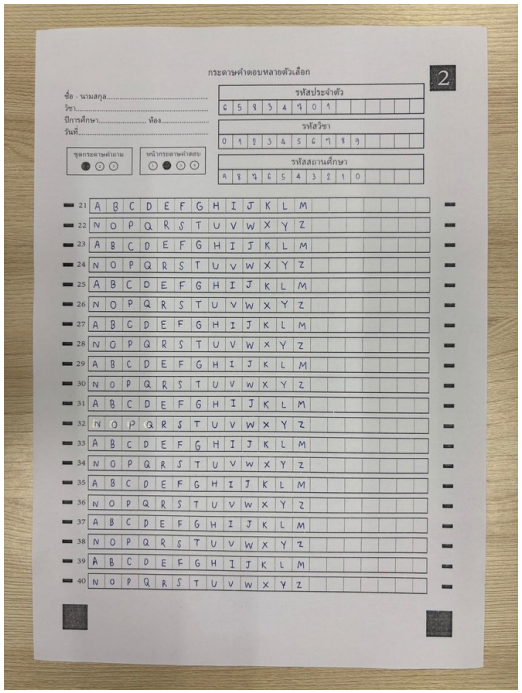
- **พัฒนาโปรแกรมตรวจข้อสอบปรนัยและอัตนัยแบบอัตโนมัติผ่านการถ่ายภาพจากมือถือ**
  โดยเริ่มจากการออกแบบหน้ากระดาษคำตอบแบบปรนัยและอัตนัยรวม 3 รูปแบบ

ตัวอย่าง Input ปรนัย

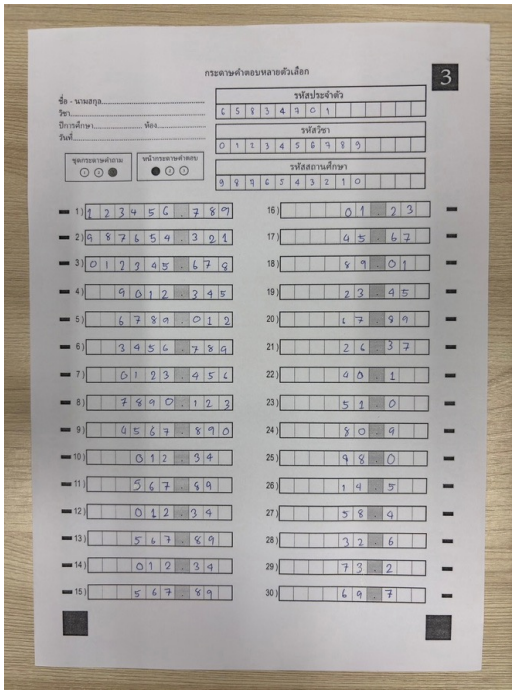


แบบฝนภาษาไทย

ตัวอย่าง Input อัตนัย

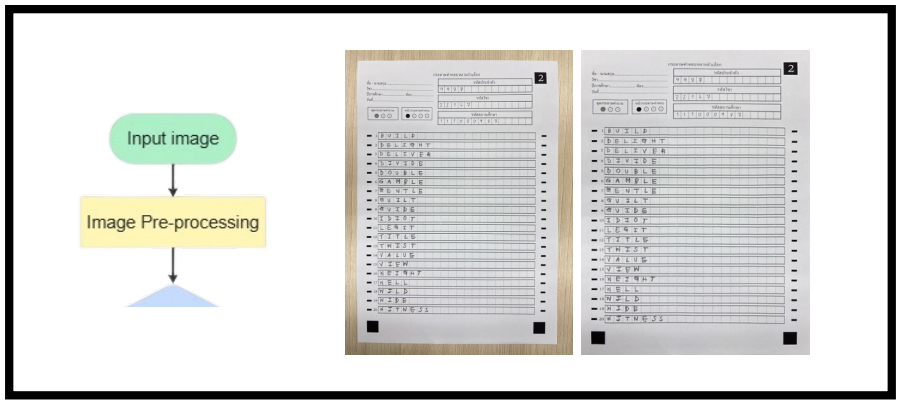


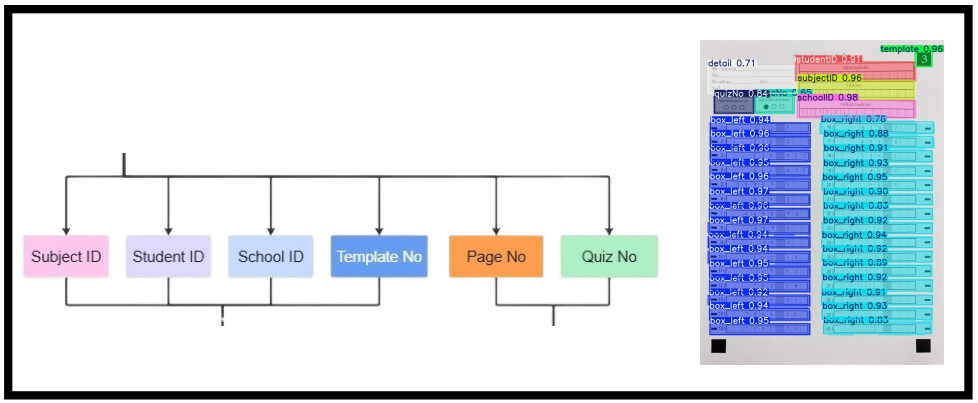
แบบภาษาอังกฤษตัวพิมพ์ใหญ่

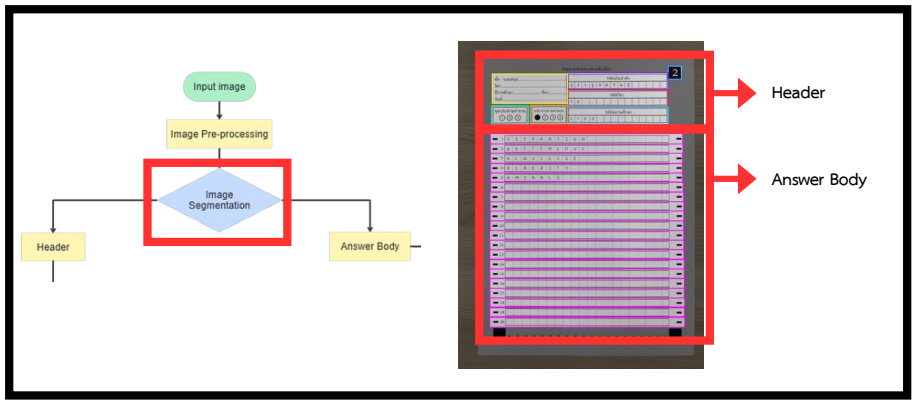


แบบตัวเลข

ตัวอย่างของ Process การทำงานของระบบ

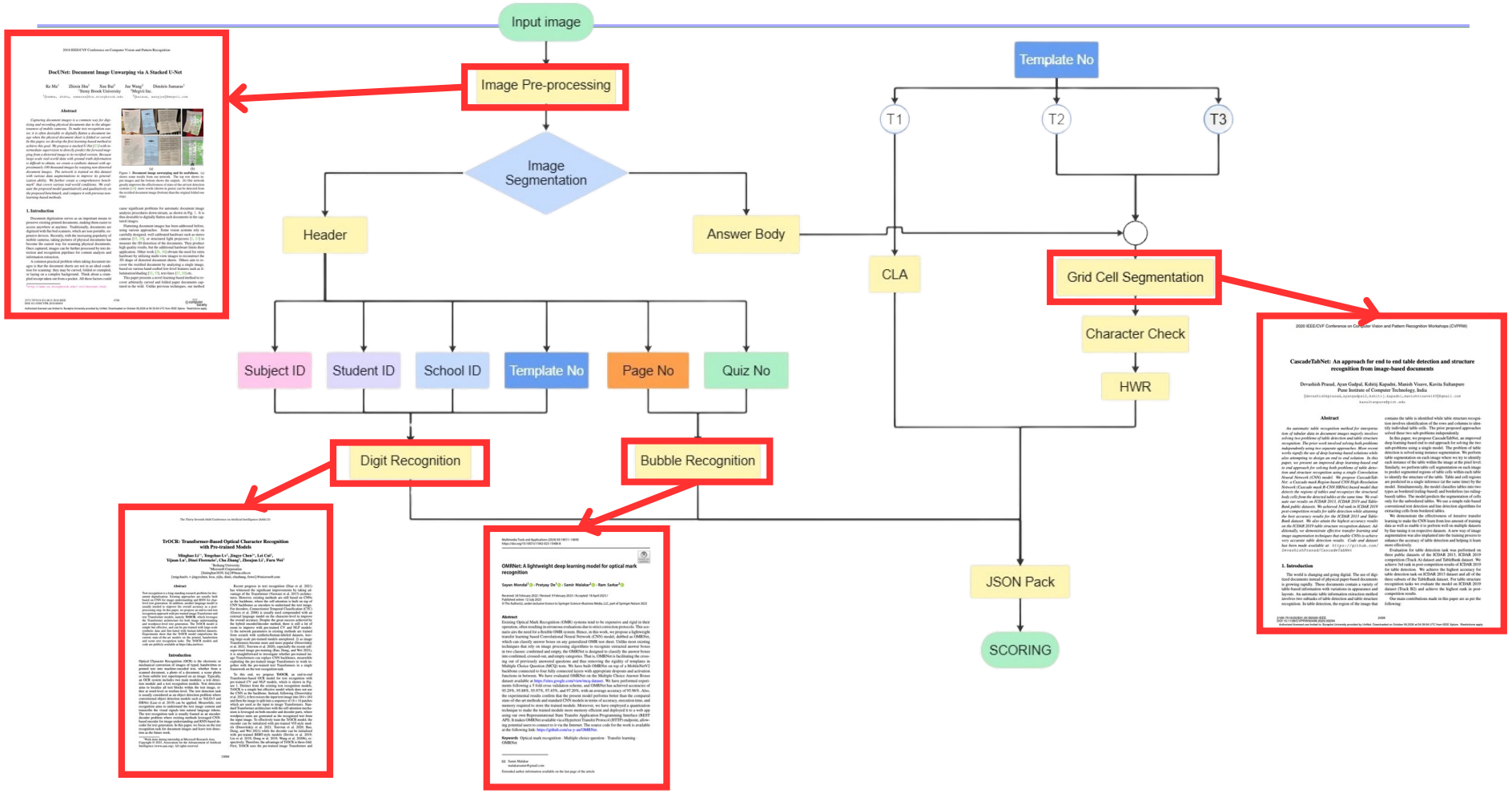








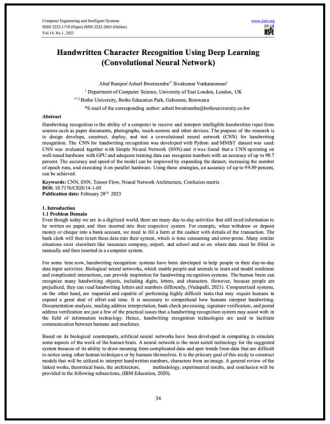
หา paper คู่เทียบและออกแบบ Flow ของระบบ

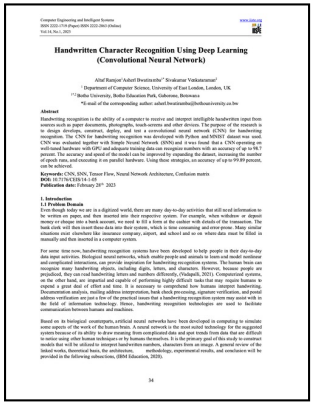


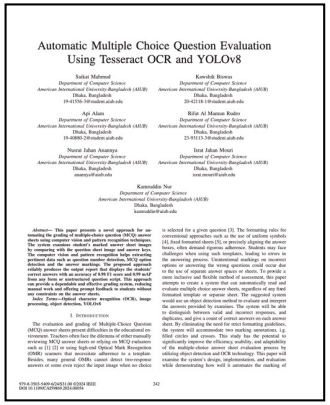
Output ที่คาดหวัง

- วัดผลตามเปเปอร์ที่เป็น Baseline ที่นำมาอ้างอิง

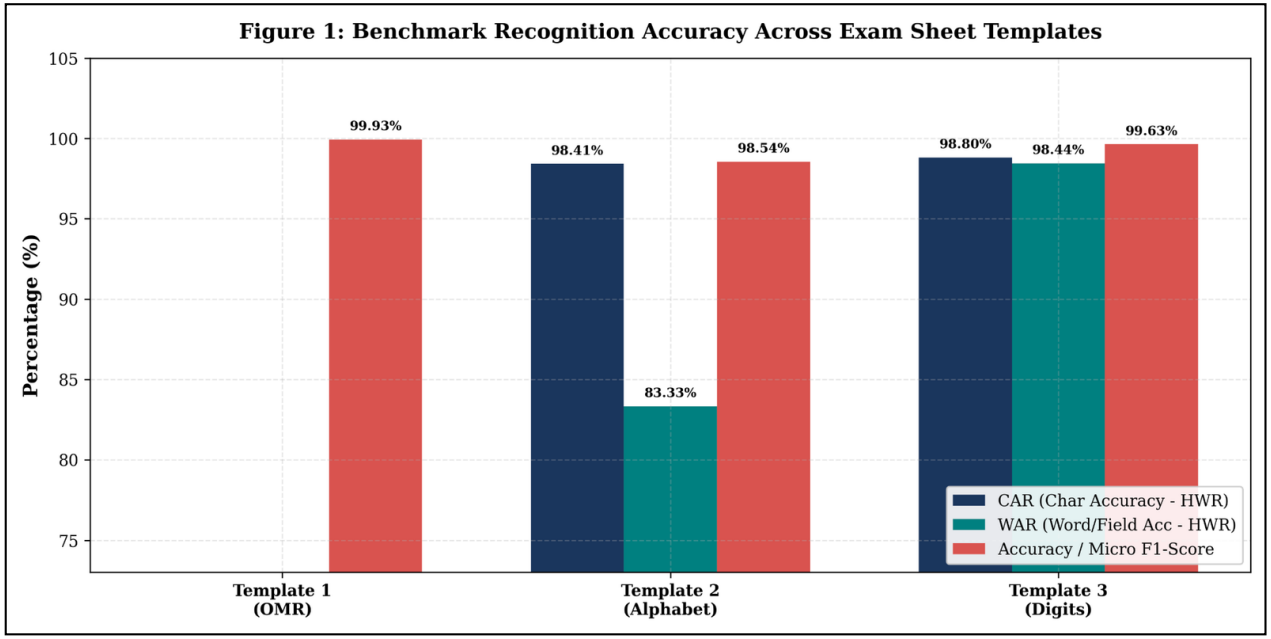

Unsupervised Optical Mark Recognition on Answer Sheets for Massive Printed Multiple-Choice Tests


Handwritten Character Recognition Using Deep Learning (Convolutional Neural Network)


Handwritten Character Recognition Using Deep Learning (Convolutional Neural Network)


Automatic Multiple Choice Question Evaluation Using Tesseract OCR and YOLOv8

ตัวอย่าง การวัดผลของ Input แต่ละแบบ



Questions ???

Q and A: What did you learn?

